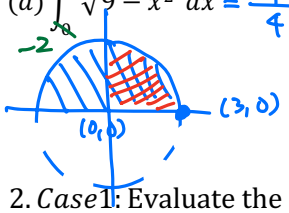
$y=\sqrt{9-x^2} \Rightarrow y^2=9-x^2 \Rightarrow x^2+y^2=9$

MAT1575, Classwork9, Fall2026

3.3 Trigonometric Substitution (p.250-259)

1. Evaluate each of the following integrals:

(a) $\int_{-2}^{3}\sqrt{9-x^2}\,dx = \frac{9}{4}\pi$



(b) $\int \frac{x}{\sqrt{9-x^2}}dx$ — let $u=9-x^2$, $du=-2x\,dx$; $\frac{du}{-2}=x\,dx$

$=\int \frac{1}{\sqrt{u}}\frac{du}{-2}$

(c) $\int x\sqrt{9-x^2}\,dx$ — $u=9-x^2$, $du=-2x\,dx$, $\frac{du}{-2}=x\,dx$

$=\int \sqrt{u}\cdot\frac{du}{-2}$

2. *Case1*: Evaluate the given indefinite integral $\int \sqrt{9-x^2}\,dx$ *Hint*: let $x=3\sin\theta$, $dx=3\cos\theta\,d\theta$

$\int\sqrt{9-9\sin^2\theta}\;3\cos\theta\,d\theta=\int\sqrt{9\cos^2\theta}\;3\cos\theta\,d\theta$

*Check the angle based on x:

$-3\le x\le 3$, $-3\le 3\sin\theta\le 3$, $-1\le\sin\theta\le 1$

$-\frac{\pi}{2}\le\theta\le\frac{\pi}{2} \Rightarrow \cos\theta\ge 0$ (*)

$=\int 3|\cos\theta|\cdot 3\cos\theta\,d\theta \overset{(*)}{=} \int 9\cos^2\theta\,d\theta=\int\frac{9}{2}+\frac{9\cos(2\theta)}{2}\,d\theta$ Under this range of θ, we have

$=\frac{9}{2}\theta+\frac{9}{2}\frac{\sin(2\theta)}{2}+C=\frac{9}{2}\theta+\frac{9}{4}\sin\theta\cdot\cos\theta+C$

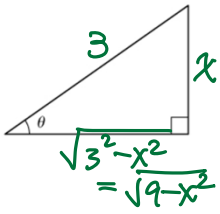
$=\frac{9}{2}\sin^{-1}\left(\frac{x}{3}\right)+\frac{9}{4}\frac{x}{3}\frac{\sqrt{9-x^2}}{3}+C$

*How to put x back to the result:

$x=3\sin\theta$, $\sin\theta=\frac{x}{3}$

$\theta=\sin^{-1}\left(\frac{x}{3}\right)$

$\cos\theta=\frac{\sqrt{9-x^2}}{3}$



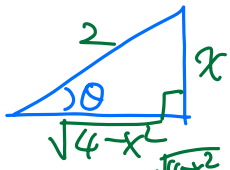
3. *Problem-Solving Strategy*: **Integrating Expressions Involving $\sqrt{a^2-x^2}$**

(1) Make the substitution $x=a\cdot\sin\theta$ and $dx=a\cdot\cos\theta\,d\theta$ which yields $\sqrt{a^2-x^2}=a\cos\theta$

(2) Simplify the expression.

(3) Evaluate the integral using techniques from the section on trigonometric integrals.

(4) Use the reference triangle to rewrite the result in terms of x: $\theta=\sin^{-1}\left(\frac{x}{3}\right)$



$\cos\theta=\frac{\sqrt{4-x^2}}{2}$, $\cot\theta=\frac{\sqrt{4-x^2}}{x}$, $\csc\theta=\frac{2}{x}$

4. Evaluate $\int\frac{\sqrt{4-x^2}}{x}dx$ $x=2\sin\theta$, $dx=2\cos\theta\,d\theta \rightarrow \sin\theta=\frac{x}{2}$

$=\int\frac{\sqrt{4-4\sin^2\theta}}{2\sin\theta}\,2\cos\theta\,d\theta=\int\frac{2\cos\theta}{2\sin\theta}\cdot 2\cos\theta\,d\theta=\int\frac{2\cos^2\theta}{\sin\theta}\,d\theta$ ($4-4\sin^2\theta=4\cos^2\theta$; $\frac{1}{\sin\theta}=\csc\theta$)

$=\int\frac{2(1-\sin^2\theta)}{\sin\theta}\,d\theta=\int\frac{2}{\sin\theta}-2\sin\theta\,d\theta=\int 2\csc\theta-2\sin\theta\,d\theta$

$=2\ln|\csc\theta-\cot\theta|+2\cos\theta+C=2\ln\left|\frac{2}{x}-\frac{\sqrt{4-x^2}}{x}\right|+2\cdot\frac{\sqrt{4-x^2}}{2}+C$

5. Evaluate $\int x^3\sqrt{1-x^2}\,dx$ two ways: (1)using $u=1-x^2$, (2)using a trigonometric substitution.

(1) $u=1-x^2$, $du=-2x\,dx$, $x^2=1-u$, $\frac{du}{-2}=x\,dx$

$\int x^3\sqrt{1-x^2}\,dx=\int x^2\sqrt{1-x^2}\,x\,dx$

$=\int(1-u)\cdot\sqrt{u}\,\frac{du}{-2}$

$=\int\frac{u^{\frac{1}{2}}-u^{\frac{3}{2}}}{-2}du=\int\frac{u^{\frac{3}{2}}}{2}-\frac{u^{\frac{1}{2}}}{2}du$

$=\frac{2}{5}\cdot\frac{u^{\frac{5}{2}}}{2}-\frac{2}{3}\cdot\frac{u^{\frac{3}{2}}}{2}+C$

$\rightarrow\frac{(1-x^2)^{\frac{5}{2}}}{5}-\frac{(1-x^2)^{\frac{3}{2}}}{3}+C$

(2) $x=\sin\theta$

$dx=\cos\theta\,d\theta$



$\sin\theta=\frac{x}{1}$, $\cos\theta=\frac{\sqrt{1-x^2}}{1}$

$\int x^3\sqrt{1-x^2}\,dx$

$=\int\sin^3\theta\cdot\cos\theta\cdot\cos\theta\,d\theta$

$=\int\sin^3\theta\cdot\cos^2\theta\,d\theta$

$=\int\sin\theta\cdot\sin^2\theta\cdot\cos^2\theta\,d\theta$

$=\int(1-\cos^2\theta)\cdot\cos^2\theta\cdot\sin\theta\,d\theta$

u-sub $=-\frac{\cos^3\theta}{3}+\frac{\cos^5\theta}{5}+C$

$=-\frac{(\sqrt{1-x^2})^3}{3}+\frac{(\sqrt{1-x^2})^5}{5}+C$

$1+\tan^2\theta=\sec^2\theta$

6. *Case2*: Evaluate the given indefinite integral $\int \frac{dx}{\sqrt{1+x^2}}$ *Hint*: let $x=\tan\theta$ $dx=\sec^2\theta\,d\theta$

*Check the angle based on x:

$-\infty<x<\infty,\ -\infty<\tan\theta<\infty,\ -\frac{\pi}{2}<\theta<\frac{\pi}{2}$

$=\int\frac{\sec^2\theta\,d\theta}{\sqrt{1+\tan^2\theta}}=\int\frac{\sec^2\theta\,d\theta}{|\sec\theta|}=\int\frac{\sec^2\theta}{\sec\theta}\,d\theta$

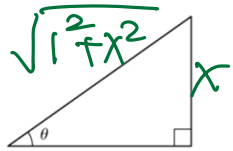
Under this range of θ, we have $\sec\theta>0$

$=\int\sec\theta\,d\theta=\ln|\sec\theta+\tan\theta|+C$

$=\ln\left|\sqrt{1+x^2}+x\right|+C$

*How to put x back to the result:

$\tan\theta=\frac{x}{1}$, $\sec\theta=\frac{\sqrt{1+x^2}}{1}$



7. *Problem-Solving Strategy*: **Integrating Expressions Involving $\sqrt{a^2+x^2}$**

(1) Make the substitution $x=a\cdot\tan\theta$ and $dx=a\cdot\sec^2\theta\,d\theta$ which yields $\sqrt{a^2+x^2}=a\cdot\sec\theta$

(2) Simplify the expression.

(3) Evaluate the integral using techniques from the section on trigonometric integrals.

(4) Use the reference triangle to rewrite the result in terms of x: $\theta=\tan^{-1}(x)$

$9\sec^2\theta-9=9\tan^2\theta$

8. *Case3*: Evaluate the given indefinite integral $\int\sqrt{x^2-9}\,dx$ *Hint*: let $x=3\sec\theta$ $dx=3\sec\theta\cdot\tan\theta\,d\theta$

*Check the angle based on x:

$x\in(-\infty,-3]\cup[3,\infty)$, $-\infty<3\sec\theta\le-3$ or $3\le3\sec\theta<\infty$

$-\infty<\sec\theta\le-1$ or $1\le\sec\theta<\infty$, $\frac{\pi}{2}<\theta\le\pi$ $0\le\theta<\frac{\pi}{2}$

$\int\sqrt{(3\sec\theta)^2-9}\;3\cdot\sec\theta\cdot\tan\theta\,d\theta$

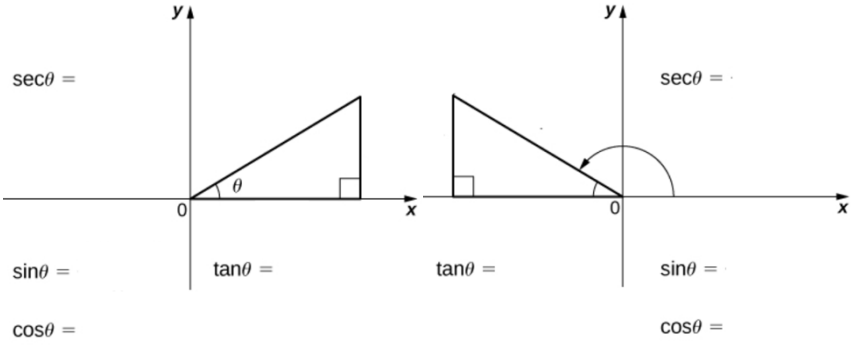
Under this range of θ, we have

$|\tan\theta|=\begin{cases}\tan\theta, & 0\le\theta<\frac{\pi}{2}\\ -\tan\theta, & \frac{\pi}{2}<\theta\le\pi\end{cases}$

$=\int 3|\tan\theta|\cdot3\sec\theta\cdot\tan\theta\,d\theta$

$0\le\theta<\frac{\pi}{2}$ $-\frac{\pi}{2}<\theta<\pi$

*How to put x back to the result:



9. *Problem-Solving Strategy*: **Integrating Expressions Involving $\sqrt{x^2-a^2}$**

(1) Make the substitution $x=$ and $dx=$ which yields $\sqrt{x^2-a^2}=$

(2) Simplify the expression.

(3) Evaluate the integral using techniques from the section on trigonometric integrals.

(4) Use the reference triangle to rewrite the result in terms of x: $\theta=$

10. Find the area of the region between the graph of $f(x)=\sqrt{x^2-9}$ and x-axis over the interval $[3,5]$.